Date

Quiz 2 - (D)

Q1

ⓐ f : $\forall x \exists y$ s.t $y.y = x$

$\sim f$: $\exists x \forall y \, (y.y \neq x)$

There is atleast one number x that doesn't have a sq. root

ⓑ f: $\exists x \exists y (x < y \wedge x^3 - x > y^3 - y)$

$\sim f$: $\forall x \forall y (x \geq y \vee x^3 - x \leq y^3 - y)$

For all pairs of nums x and y, either x is grester than or equal to y or $x^3 - x$ is less than or equal to $y^3 - y$.

ⓒ f: $\forall x \forall y \left[(x^3 + x - 2 = y^3 + y - 2) \rightarrow (x = y) \right]$

$\sim f$: $\exists x \exists y \left[(x^3 + x - 2 = y^3 + y - 2) \wedge x \neq y \right]$

There exist two diff. values x and y where $x^3 + x - 2$ is equal to $y^3 + y - 2$ while x is not equal to y.

Q2

ⓐ All students at the school have seen atleast one movie.

ⓑ There exist two diff. students at the school s.t if the first one has seen a movie then the second one has also seen it.

Q3

ⓐ $\forall x \left[(S(x) \wedge C(x)) \longrightarrow \exists y \, (S(y) \wedge H(x,y)) \right]$

ⓑ $\exists x \left[S(x) \wedge \forall y (S(y) \rightarrow H(x,y)) \right]$

ⓒ $\exists x \left[S(x) \wedge \forall y ((S(y) \wedge C(y)) \longrightarrow \sim H(x,y)) \right]$

ⓓ No;

$\forall x \, (C(x) \rightarrow S(x))$ is an if-then statement, it does not go both ways. To assume it does would be the converse error.